

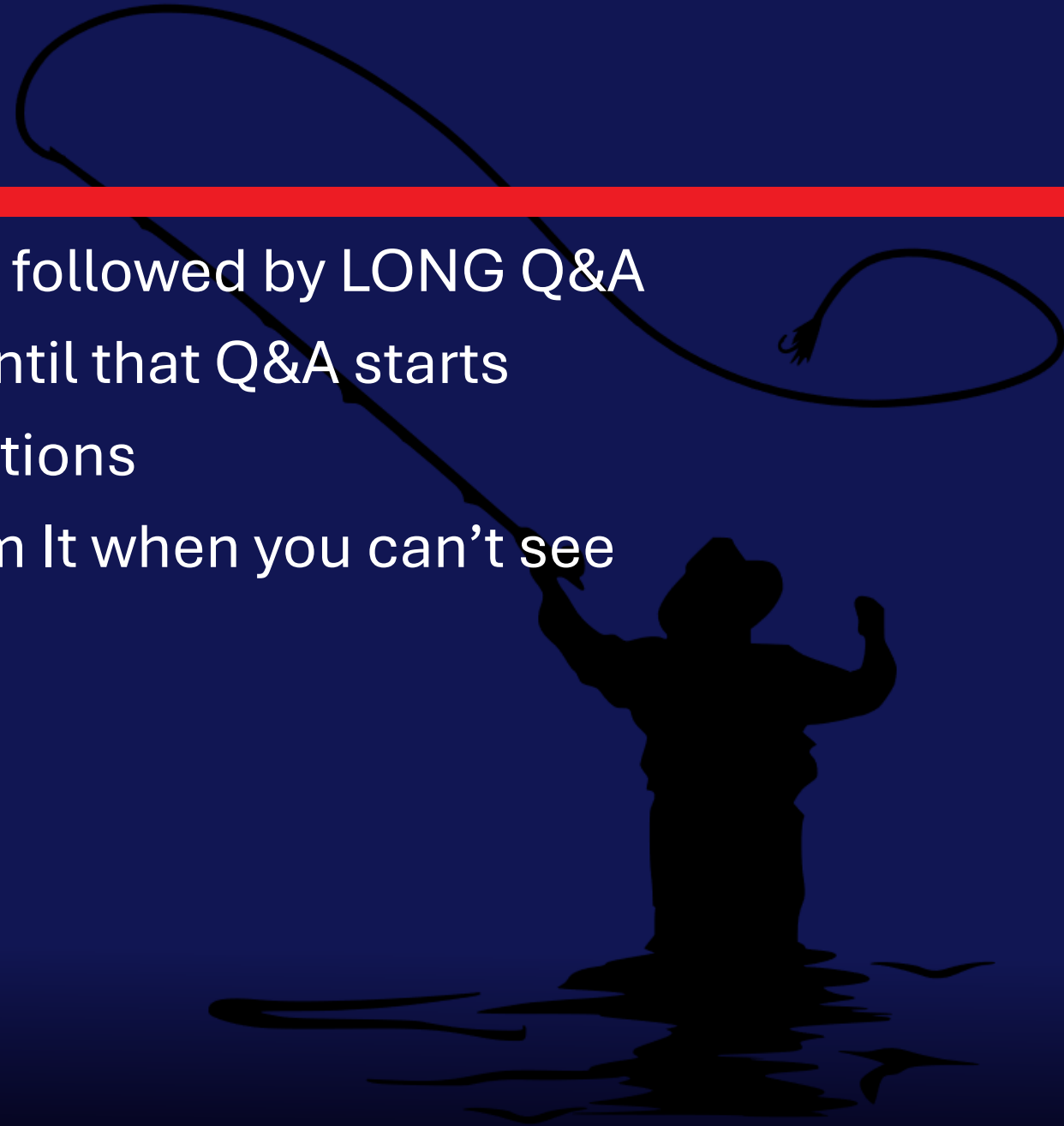
Zero Trust – The Great Debate

Zero Trust – Dope or Nope?



Attention

- ◆ MMS sessions are 60-75 minutes followed by LONG Q&A
- ◆ Please hold detailed questions until that Q&A starts
- ◆ Feel free to ask clarification questions
- ◆ Remind the speakers to use Zoom It when you can't see



Sami Laiho

Chief Research Officer / MVP

- IT Admin since 1995 / MCT since 2001
- Microsoft MVP in Windows OS since 2011, Security since 2024
- **"100 Most Influential people in IT in Finland" – TiVi'2019 →**
 - **Among Top 10 most followed techies on X/Twitter in Finland**
- Specializes in and trains:
 - Windows Internals
 - Security, Social Engineering, Auditing
- Trophies:
 - Best Session at Advanced Threat Summit 2020
 - Best Speaker at NIC, Oslo 2016, 2017, 2019, 2020, 2022, 2023, 2024 & 2025
 - **Ignite 2018 – Session #1 and #2 (out of 1708) !**
 - TechEd Europe and North America 2014 - Best session, Best speaker
 - TechEd Australia 2013 - Best session, Best speaker





X (ex-Twitter): @samilaiho
Bluesky: @samilaiho.com
LinkedIn !



Sami Laiho  @samilaiho · 5h

Every security person out there, please:

Buy the Concept, Not the Product

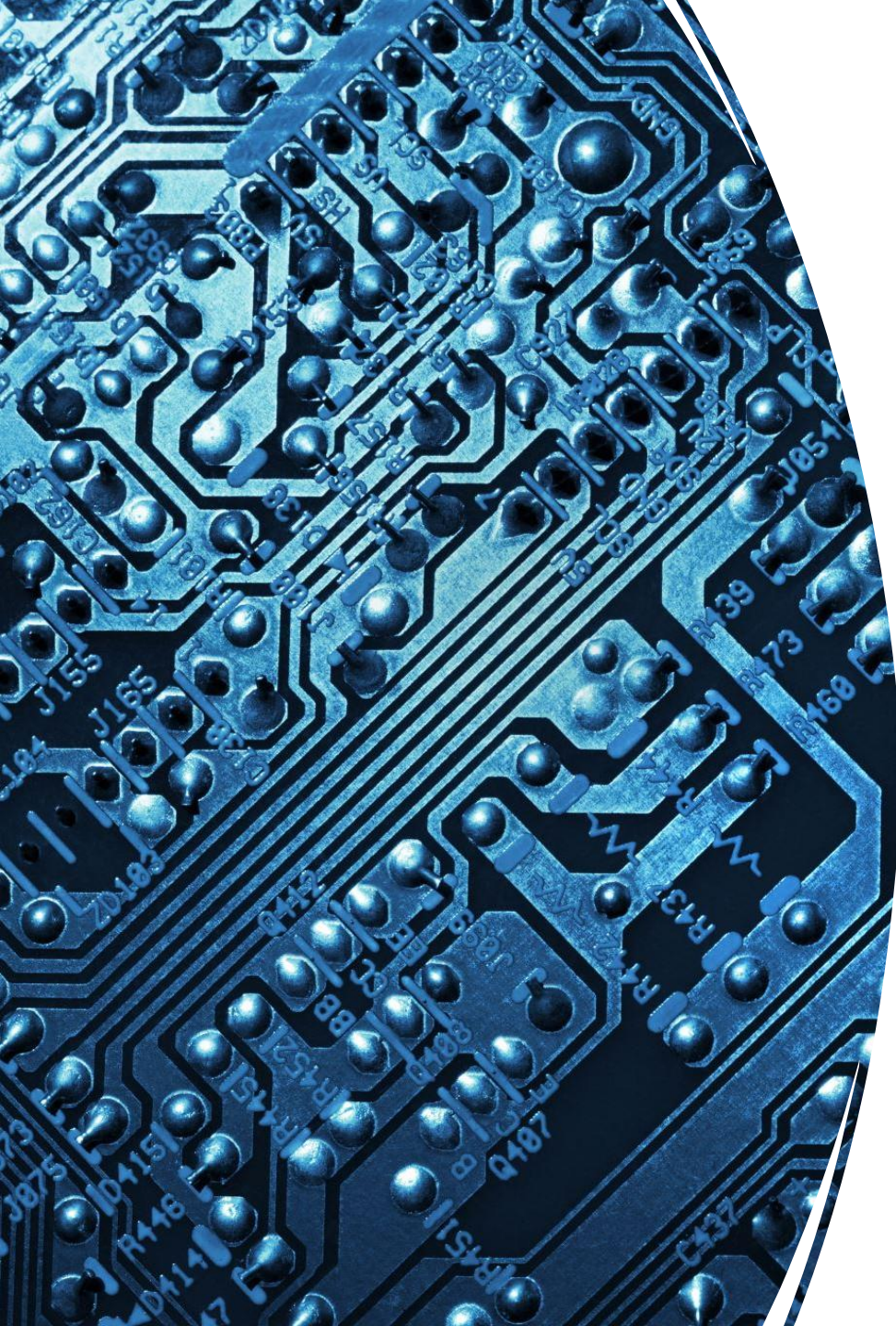
“We want EPM but for that they say we need to get rid of admin rights”

A real quote from a customer





Zero Trust Principles



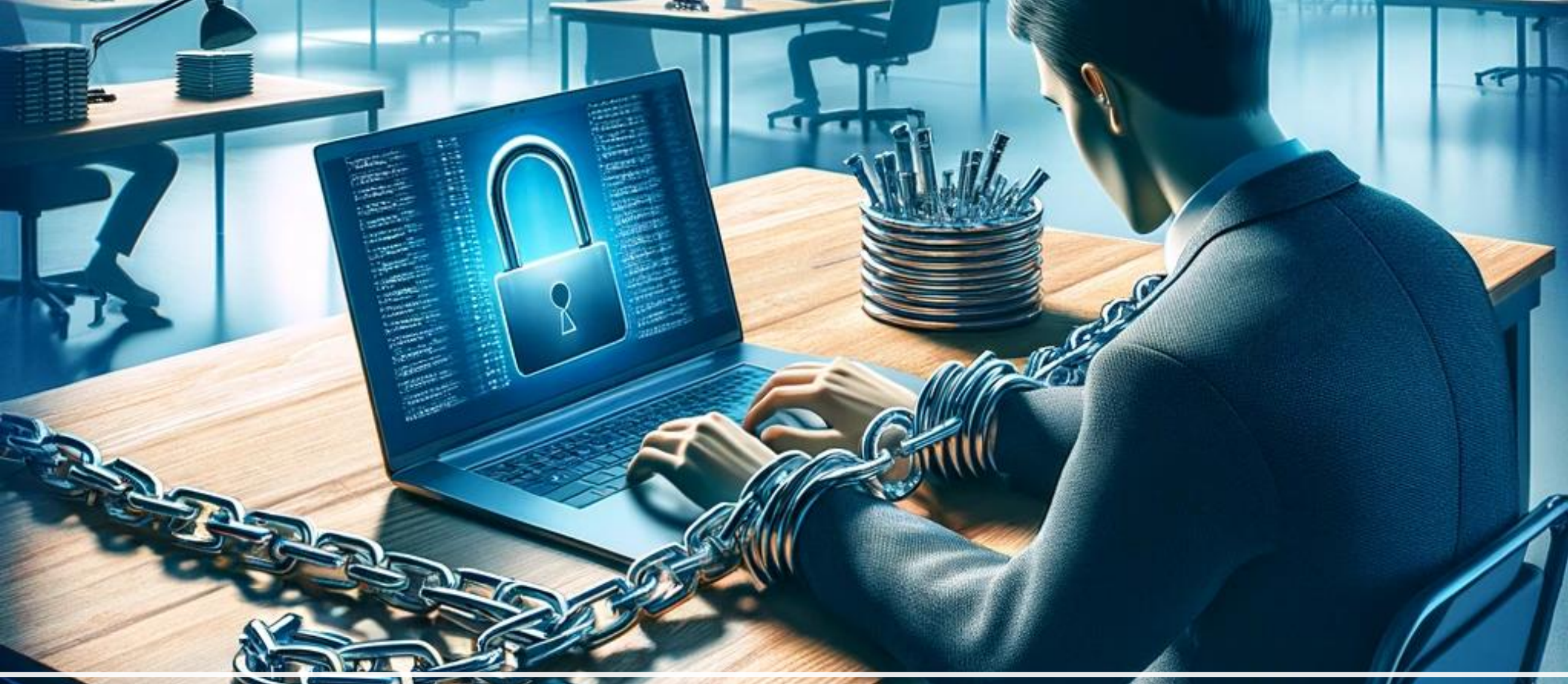
In April 1994, the term "zero trust" was coined by Stephen Paul Marsh

- Wikipedia: There are several ways to implement all the tenets of ZT; a full ZTA solution will include elements of all three:
 - Using enhanced identity governance and policy-based access controls.
 - Using micro-segmentation
 - Using overlay networks and software-defined perimeters
- NCSC
 - Single strong source of user identity
 - User authentication
 - Machine authentication
 - Additional context, such as policy compliance and device health
 - Authorization policies to access an application
 - Access control policies within an application

Cyber Security **BULLSHIT BINGO**

"Hope my demo works."	"That is supposed to work."	"The era of AI."	"You can find me on twitter."
"Thanks to the sponsors."	"Azure AD."	"Let's talk about that offline."	"Market Leader."
"That's a feature."	"This is grown historically."	"Zero Trust."	"Hope I have some SCU left."
"AI Transformation."	"Best of breed versus best of suite."	"GenAI."	"That's great feedback"

We make IT security a defensive art.



ZeroTrust is the worst name ever...



Zero Trust principles



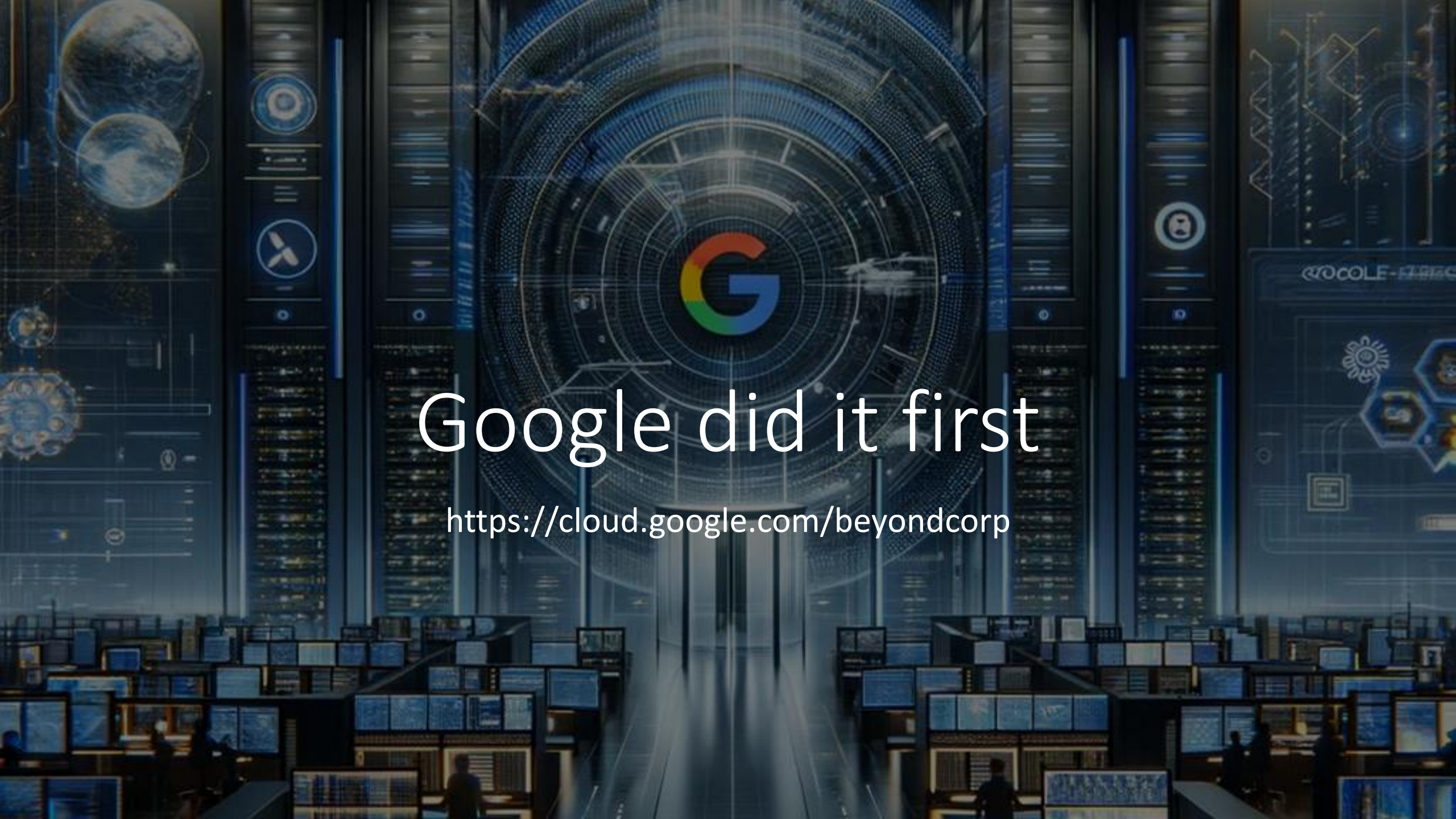
Verify explicitly



Use least privileged access



Assume breach



Google did it first

<https://cloud.google.com/beyondcorp>



MFA

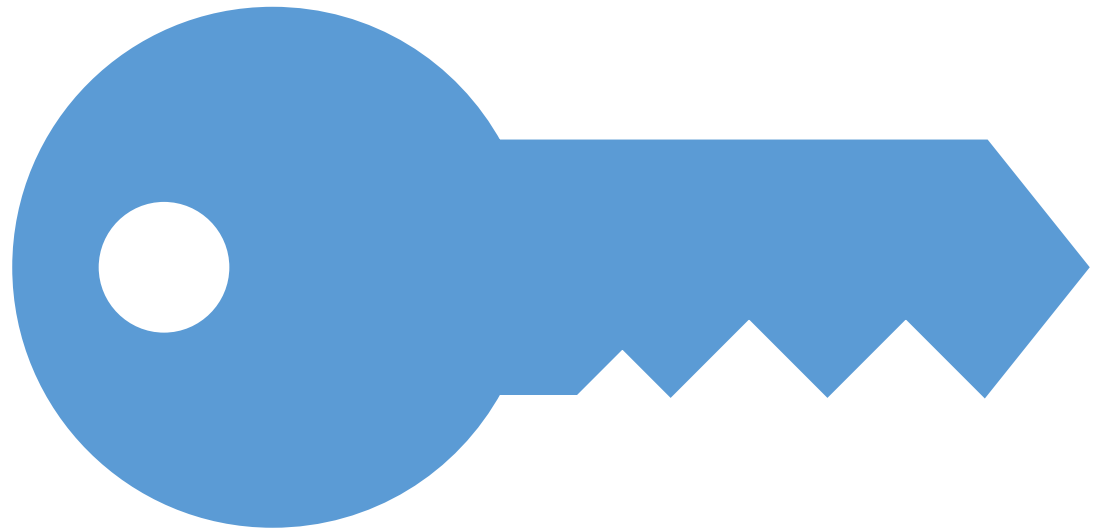
Verify Explicitly



MFA

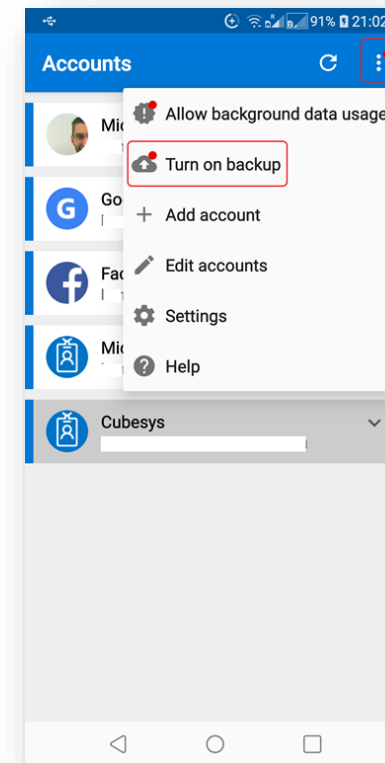
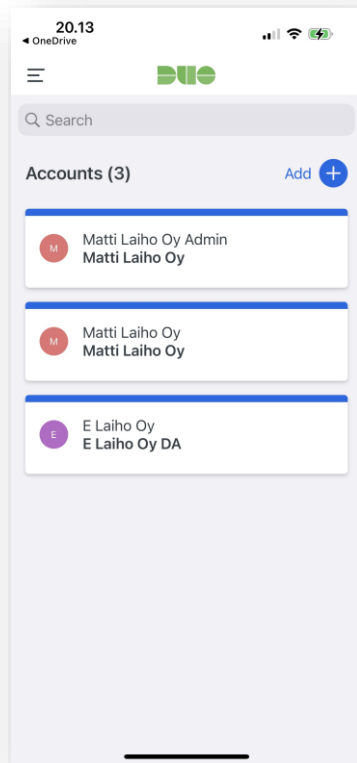
- Always verify → MFA everywhere
- Has changed the most used attack vector from Phishing → Unpatched Vulnerabilities
- Can be made accessible and easy enough to use
- Passwordless / Passkeys are the future but we can't wait

Factors



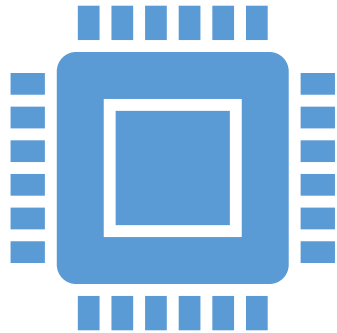
Authenticator Apps?

Requires a Smartphone



ADMINIMIZE
ADMINIMIZE

PIN-codes



Windows with PIN is MFA!



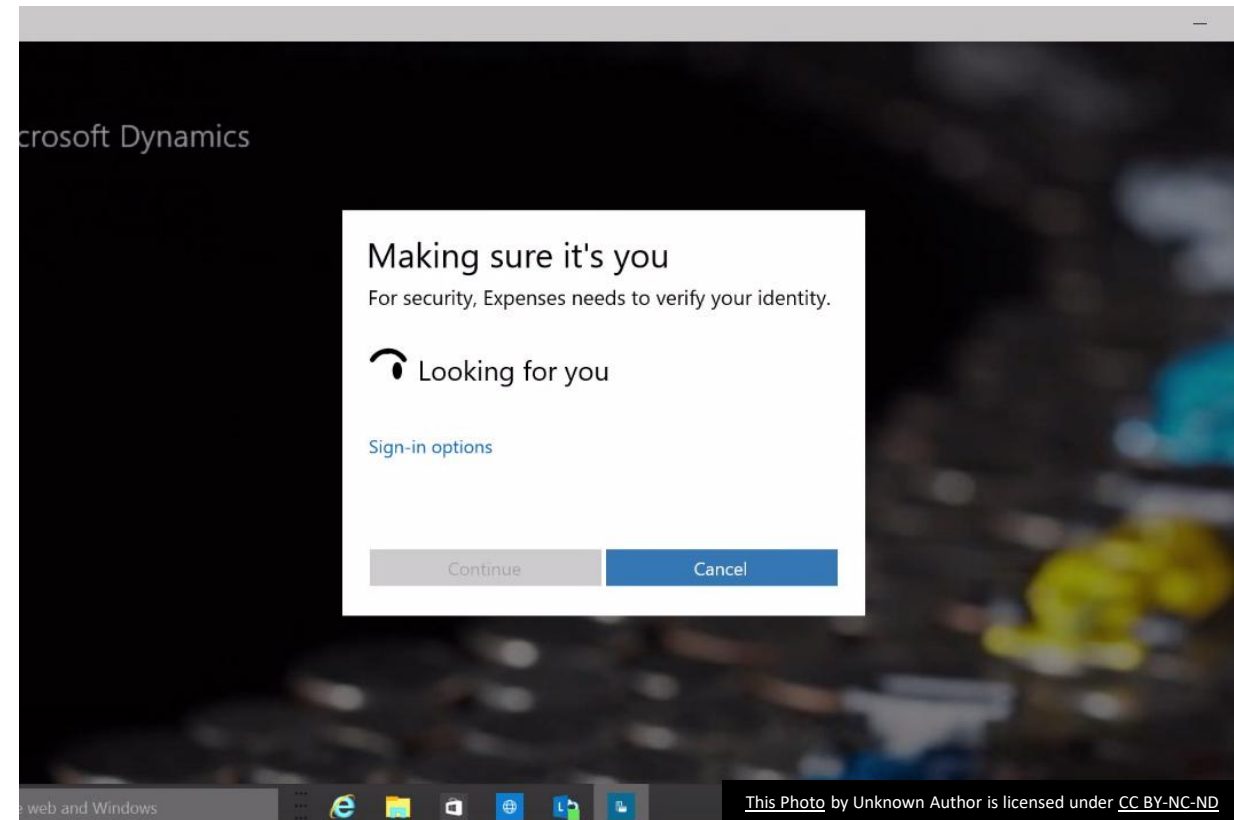
Why a PIN is more secure than a
password



Biometrics



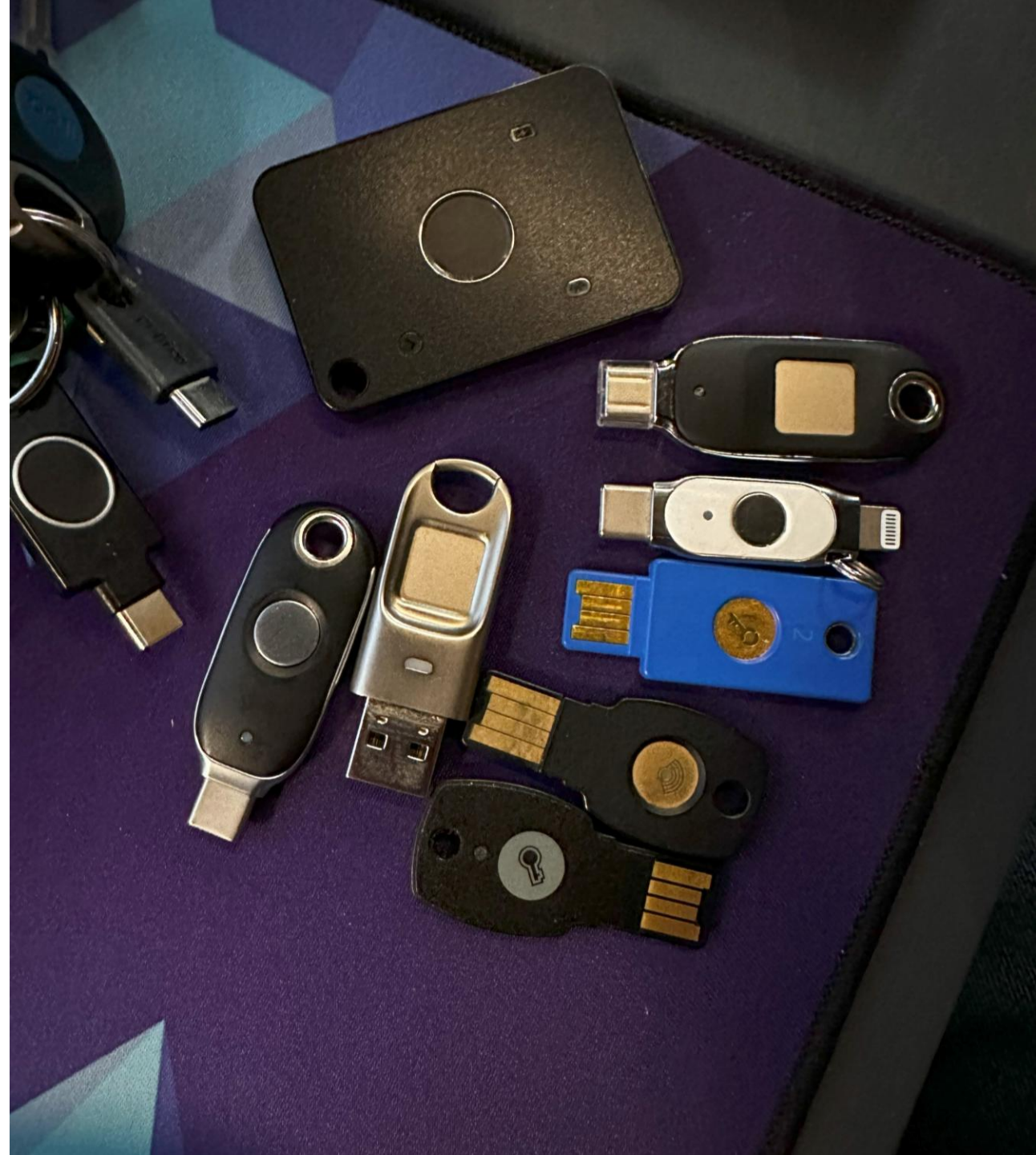
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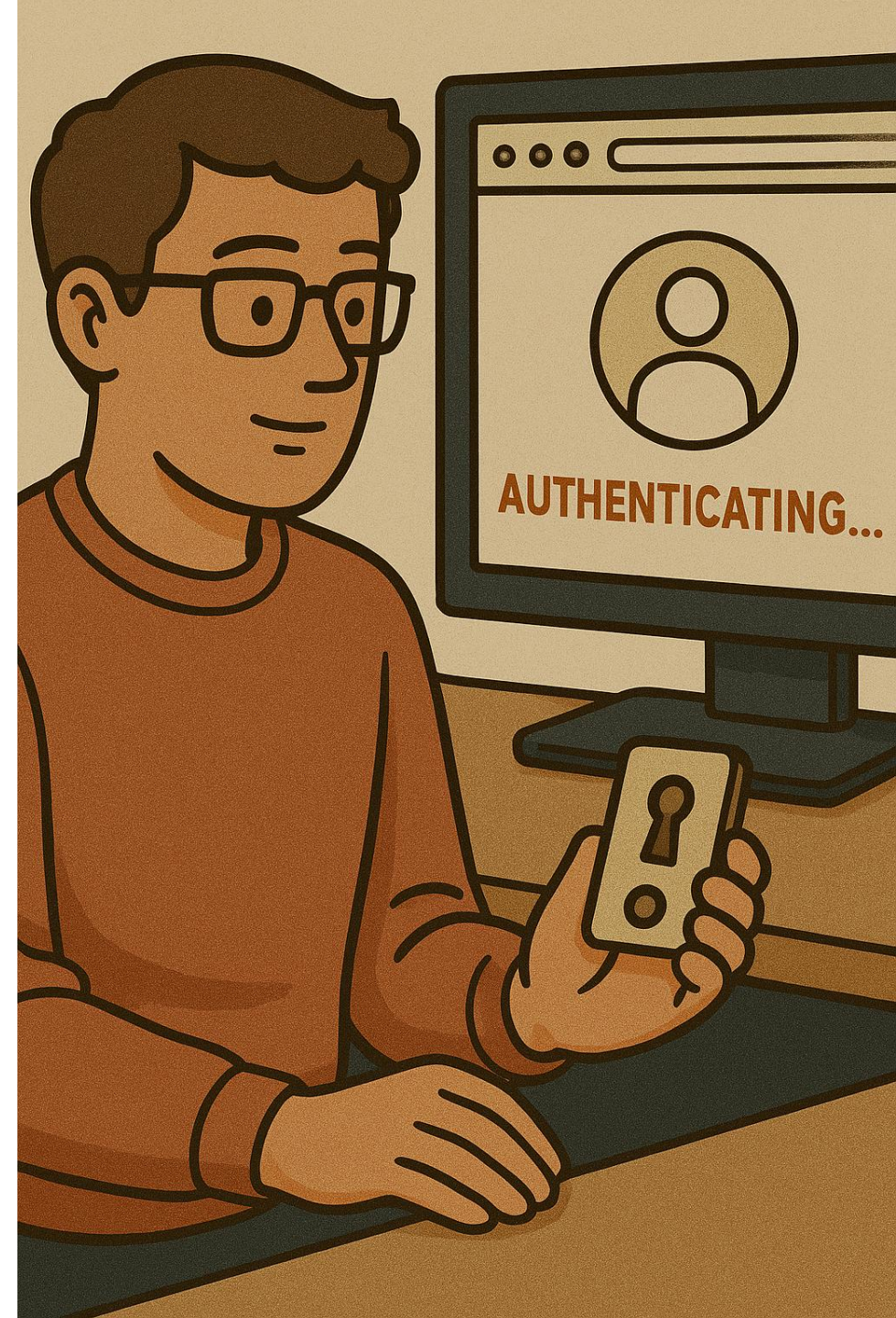
Physical Tokens

- Smartcards?
- Phishing resistant MFA
 - Plug in or NFC
 - Biometrics or PIN



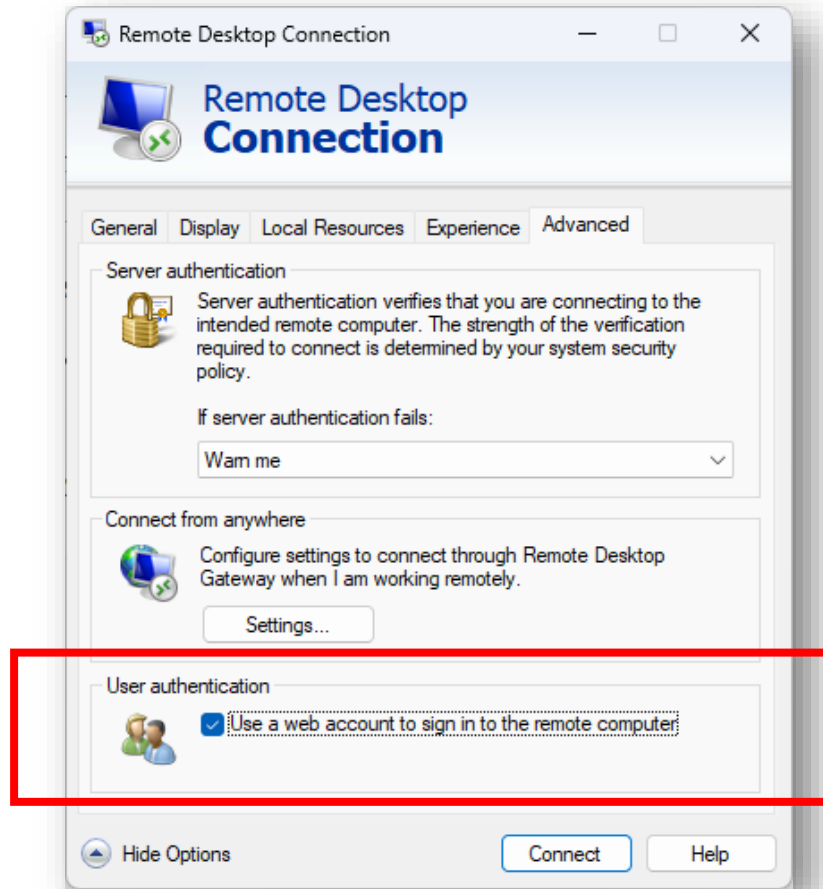
Combined with PassKeys = Muy Bien!

Now even “roaming” with Synced keys in preview!



Yubikeys

<https://swjm.blog/the-complete-guide-to-rdp-with-yubikeys-fido2-cba-1bfc50f39b43>





Least Privilege

NT 3.1 Security Guide



States that local admins have full access to the computer. FULL STOP.

Administrators Properties

General



Administrators

Description:

Administrators have complete and unrestricted access to the computer/domain

Critical risks of Administrative Rights



The risk increases dramatically from loss of one user's assets, to losing the whole company operations.



It allows the malicious tools to operate on Windows deeper and at hardware levels.



It prevents the company from controlling the computer settings and data, therefore managing risk.



It allows for identity theft.



Shadow IT is rampant as the end user controls the device



The Principle of Least Privilege is a Core Component of Zero Trust

Why Shouldn't Users Want to Have Administrative Rights?

Performance of systems is improved without local administrator rights

Installation of various software causes slowdowns and reliability issues.

SSD lifespan increases

Decreases the need for reinstallations.

Due to more reliable systems productivity is increased

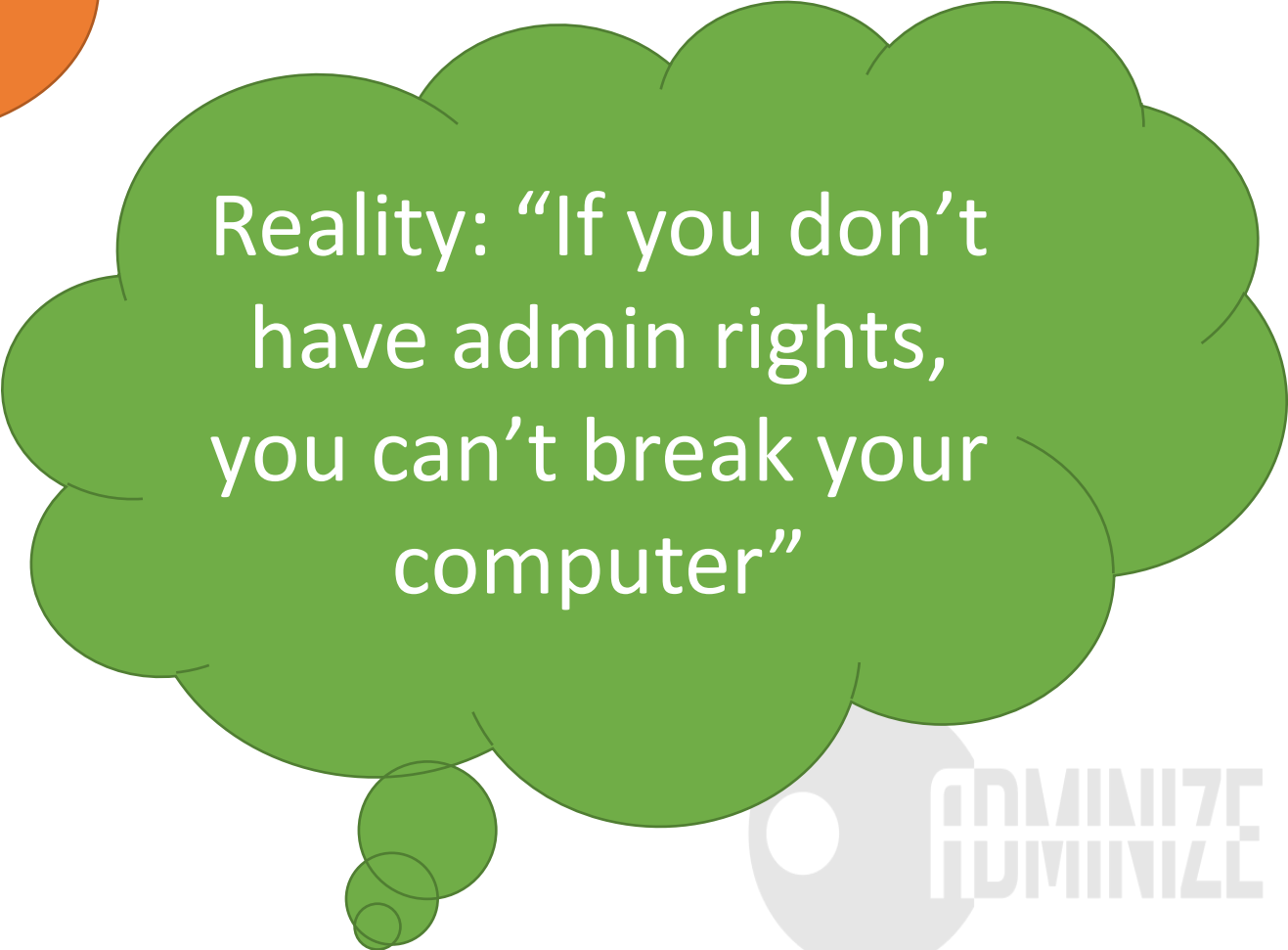
The attack surface of malware is greatly reduced and helps limit propagation

Because it decreases the amount of money needed in extra security solutions

Because it mitigates more vulnerabilities than patching



Users: “If I don’t have
admin rights, I can’t
fix my computer”



Reality: “If you don’t
have admin rights,
you can’t break your
computer”

Admin Rights are not Human Rights

- US:
 - <https://www.zazzle.com/store/adminize>
- EU:
 - <https://www.zazzle.co.uk/store/adminize>



Stop end-users from hurting their computers

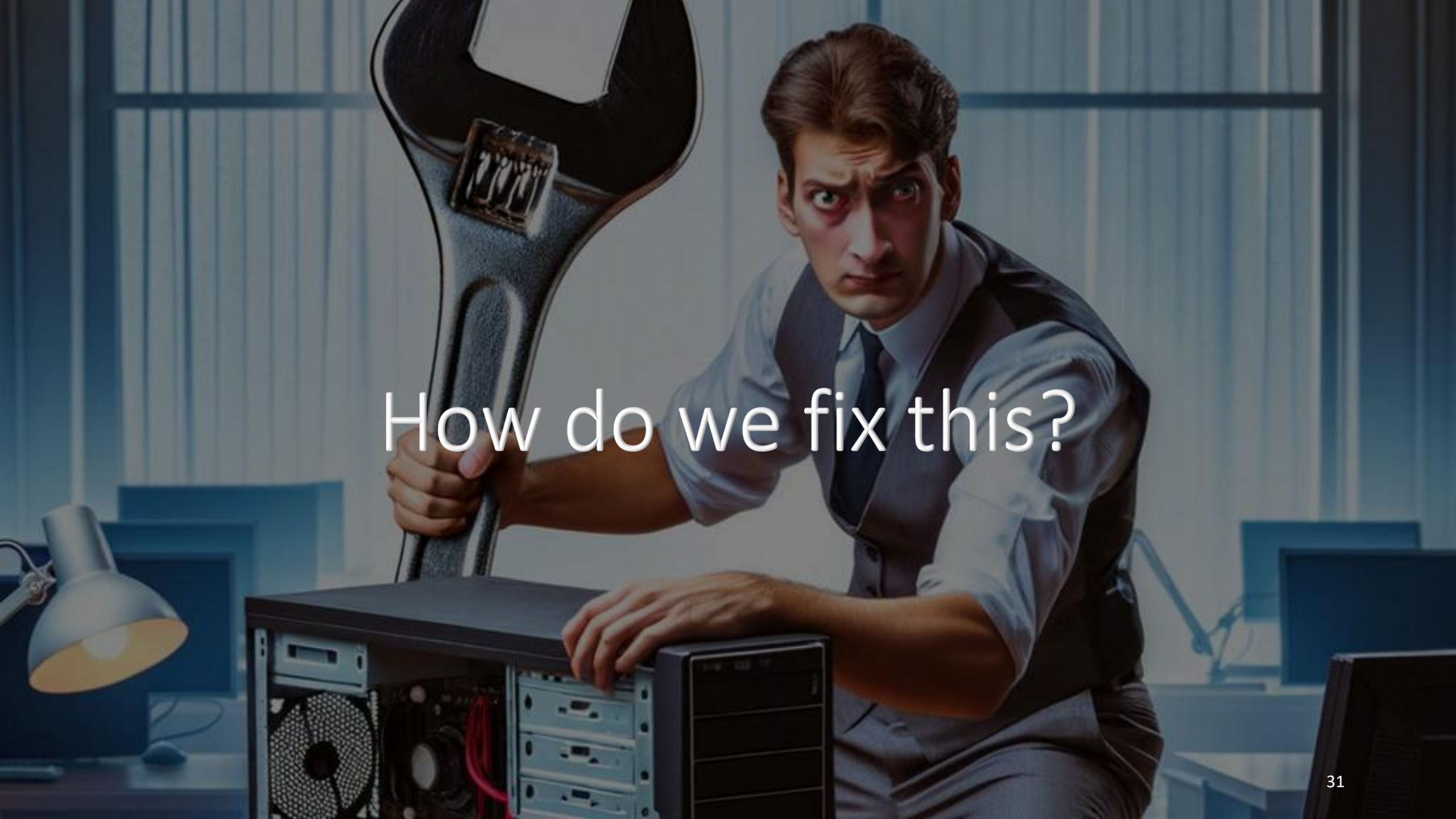
Jannik ~~XXXXXXXXXX~~

we had a decrease of 75% of
tickets after we implemented no
admin rights and acevto.... so
price to us is crazy cheap



A large office with many people working at computers. In the foreground, two men are sitting at a desk, playing ping-pong on a small table. The desk is cluttered with computer equipment, including monitors, keyboards, and a large stack of server units underneath. The background shows other employees working at their desks, some holding ping-pong paddles. The office has a grid ceiling with fluorescent lights and various charts and papers on the walls.

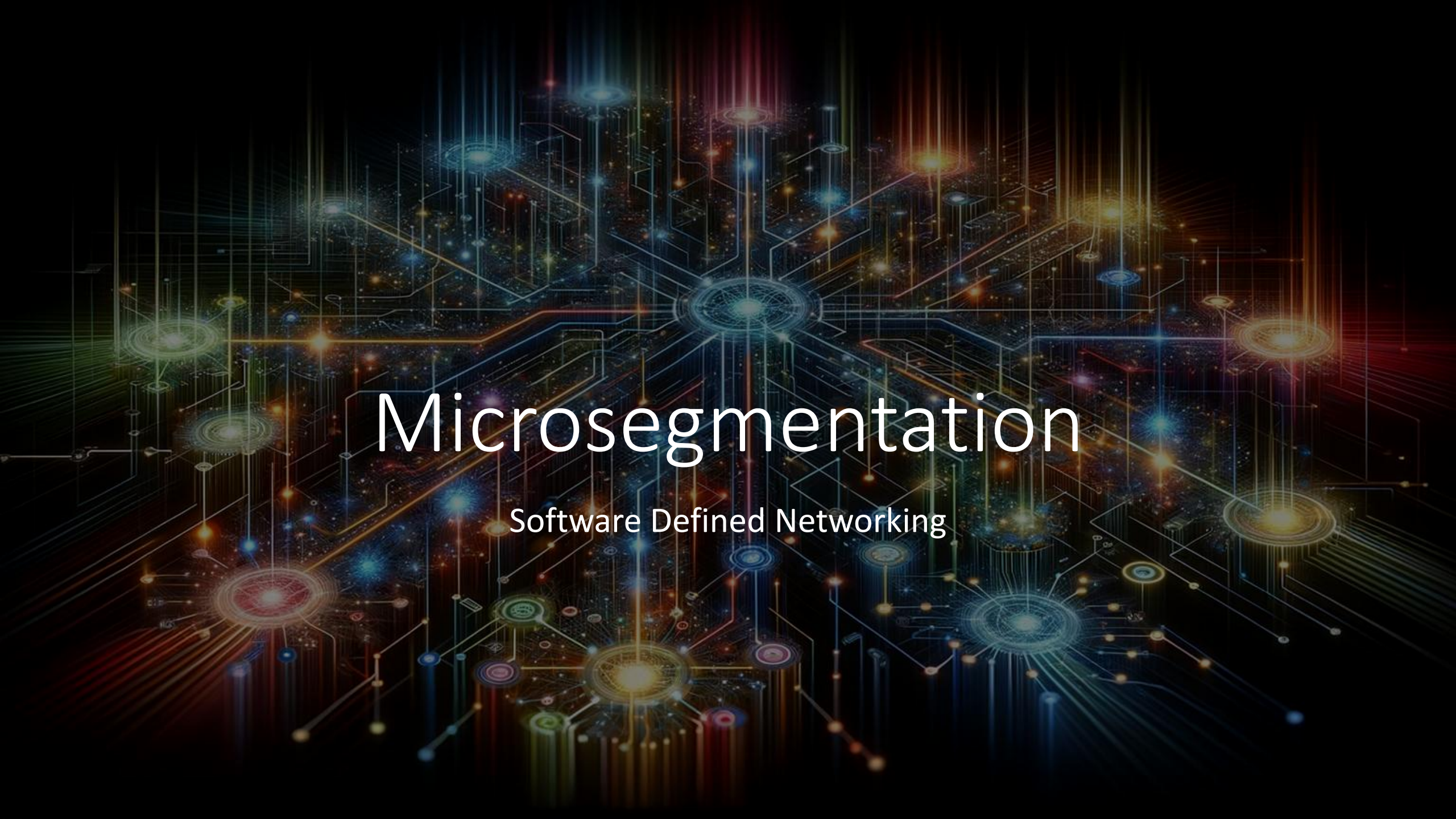
US Customer: 65% less
reinstallations

A man in a light blue shirt and dark vest is leaning over a server rack in a data center. He is holding a large, oversized wrench over the rack. He has a concerned expression on his face. The background shows other server racks and a desk lamp.

How do we fix this?

You Need an EPM!

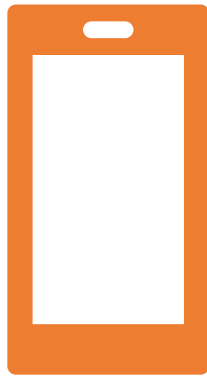
Endpoint Privileged Management (I don't mean Microsoft EPM)

The background is a complex, abstract digital network. It features a dense web of glowing lines in various colors (blue, green, yellow, orange, red) that connect numerous circular nodes. Some nodes are larger and more prominent, while others are smaller. The overall effect is a sense of a vast, interconnected digital space with a futuristic, high-tech aesthetic.

Microsegmentation

Software Defined Networking

Micro-Segmentation



Controlling flow between every
node or every app



Controlling “East-West”, instead
of “North-South”

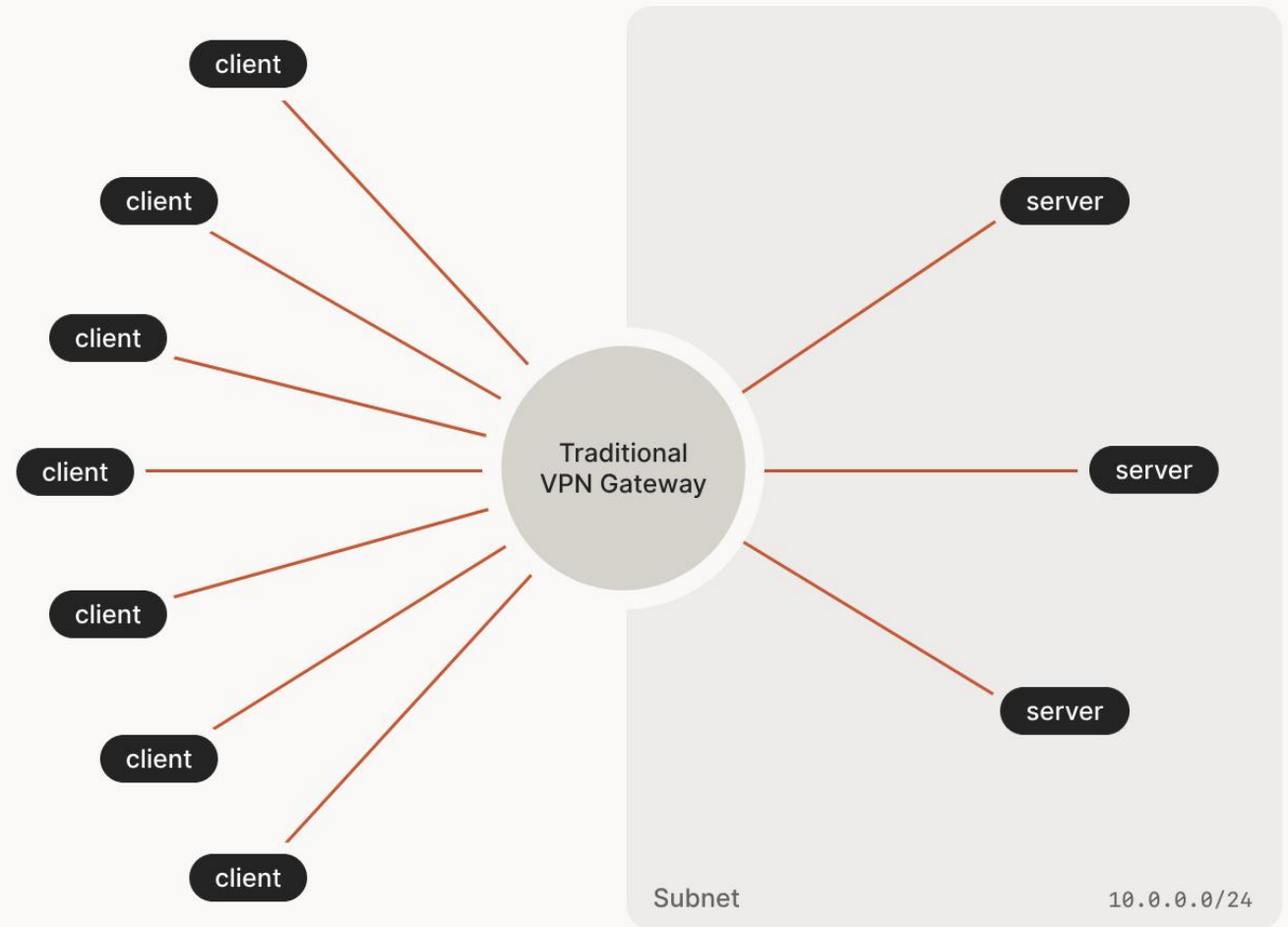


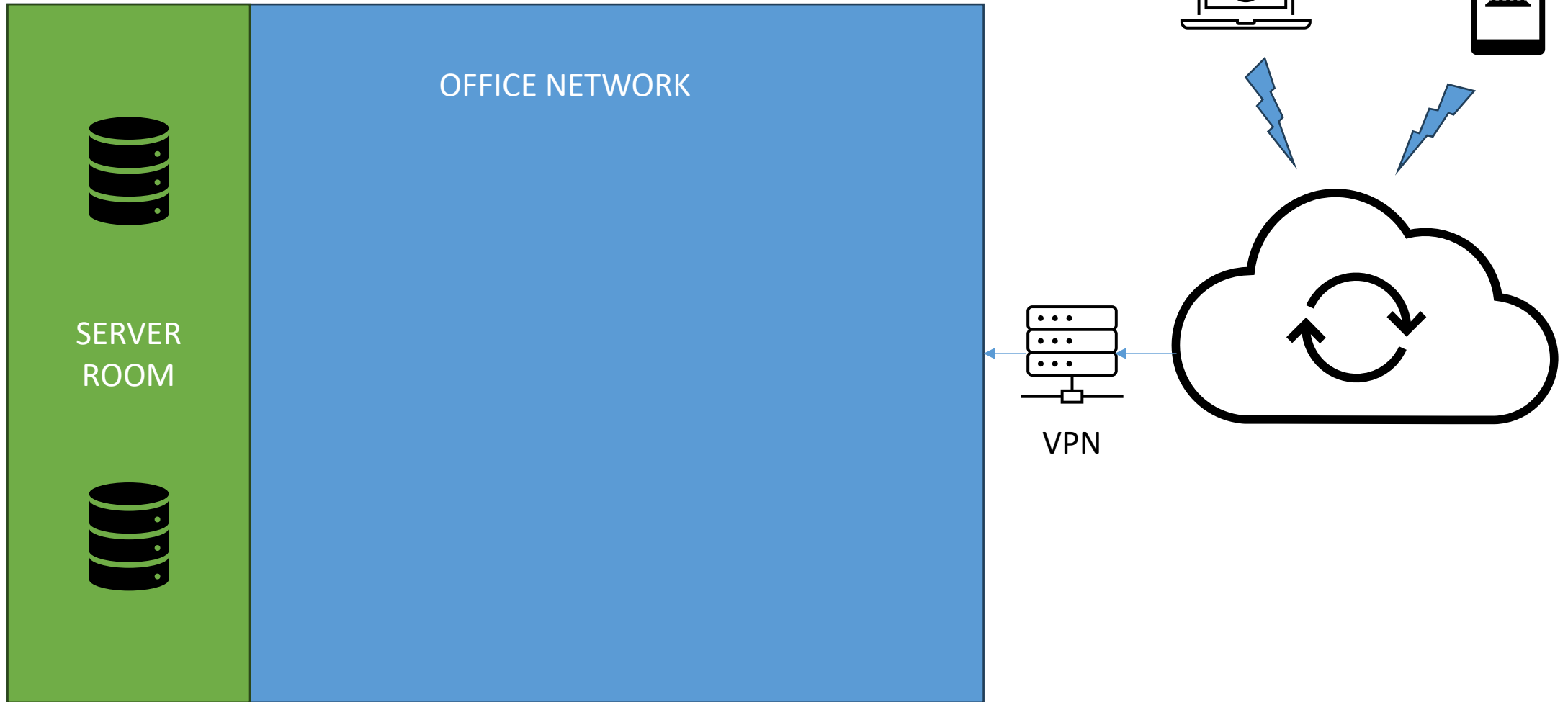
What are we Being Offered?

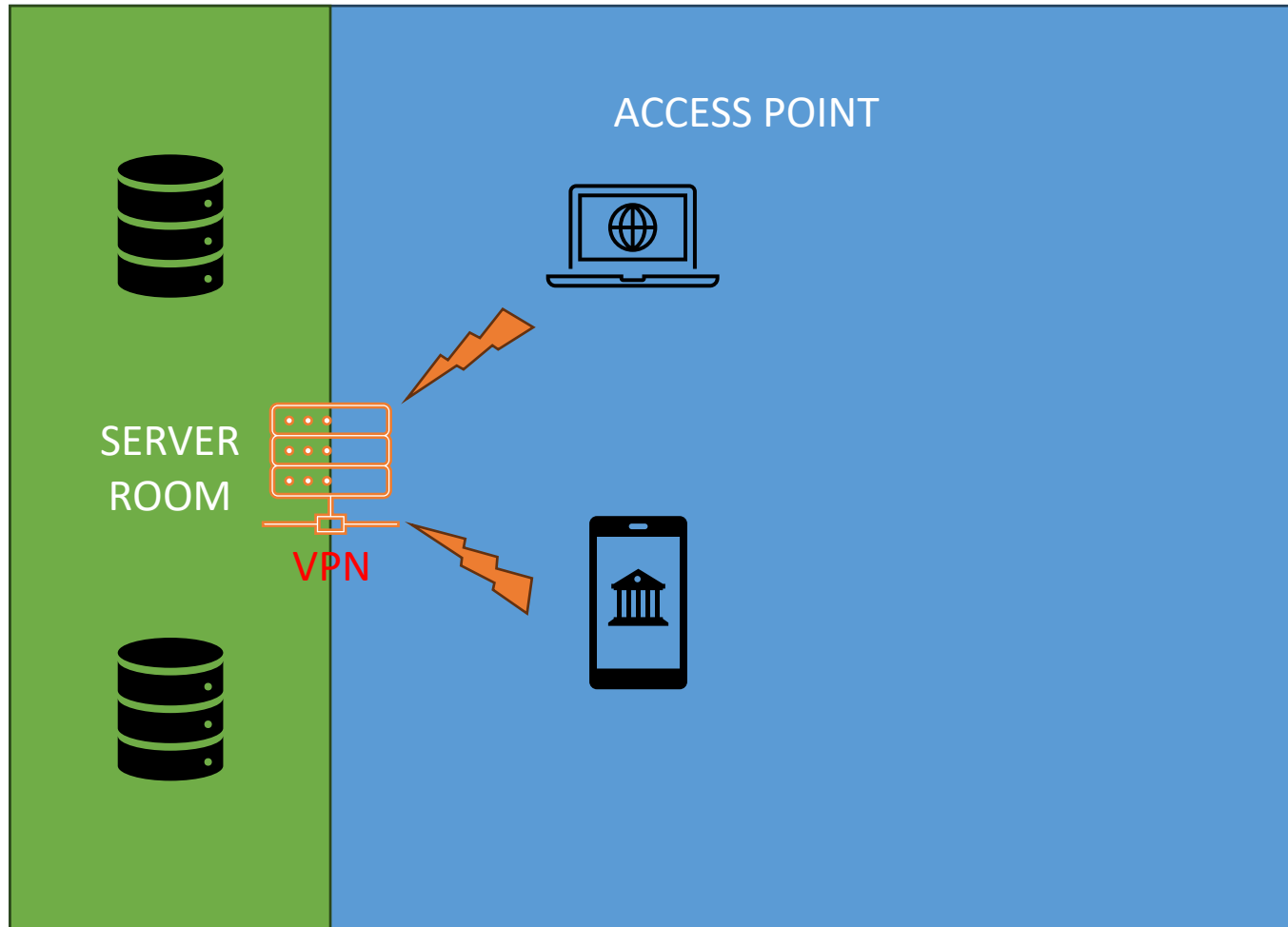
Move the VPN-
border



Traditional VPN

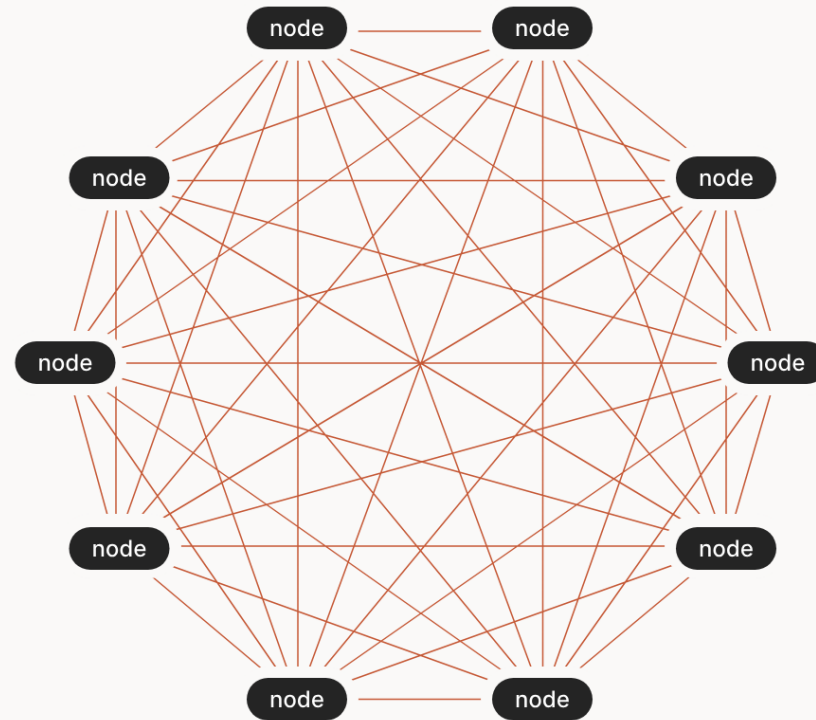






WireGuard

MESH



$n(n-1) = 90$ WireGuard endpoints (for 45 connections)

A group of about ten children are gathered around a large wooden table in a dimly lit room, playing video games. They are using multiple large monitors and keyboards. The room is decorated with various posters on the wall, including one of a robot, one of a character in a suit, and one of a character with a sword. There are also several small figurines and toys on the table and shelves. A large window with sheer curtains is visible on the right side of the room. The overall atmosphere is one of intense gaming activity.

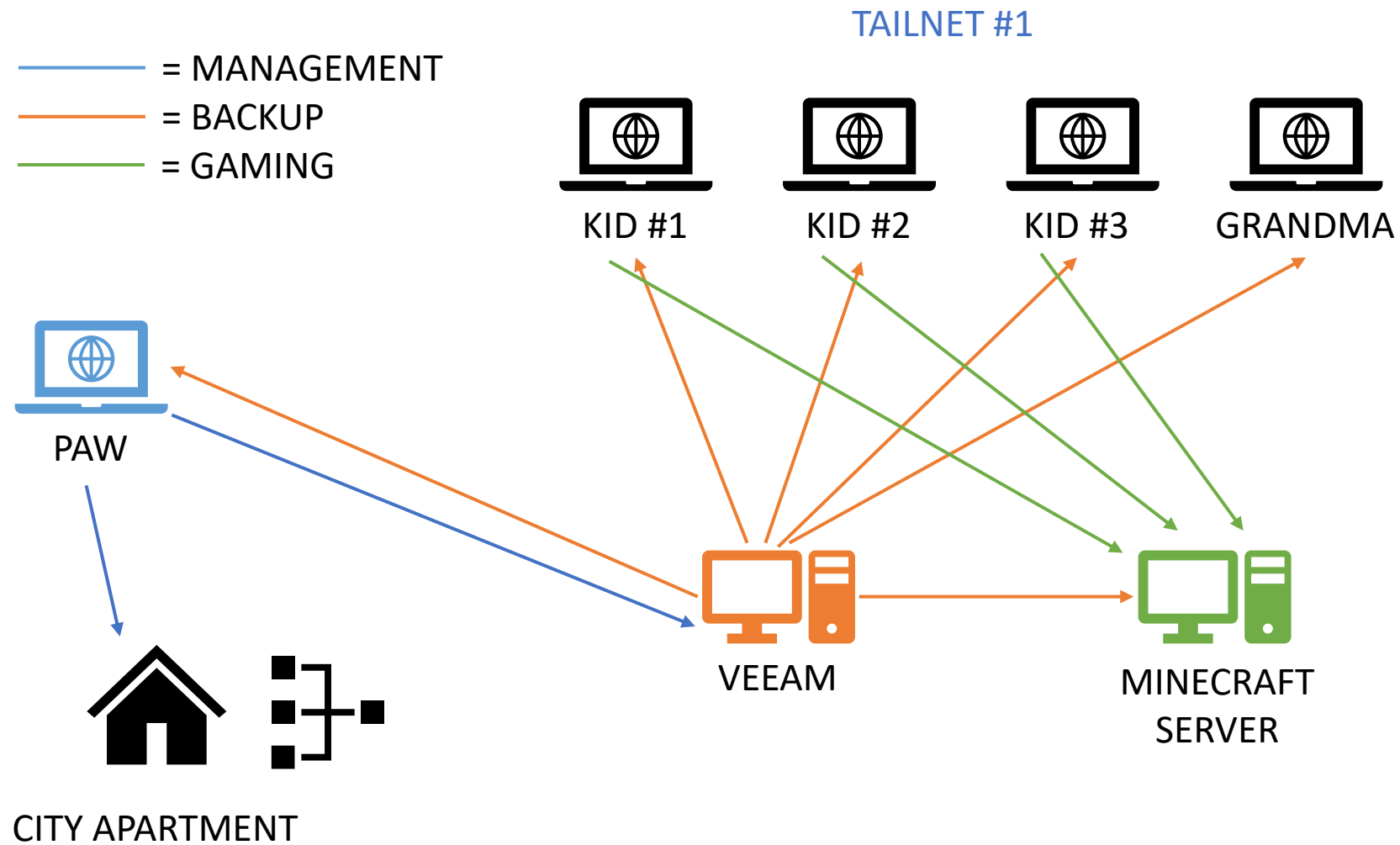
Zero Trust with my Children

Just proves how bad the name is...



What has an
SLA of 3.67s ?





Now the Minecraft-issue...

Out of my datacenter, out of my country, roaming iPhone, no keyboard, no mouse, no screen, pissed off children... → Demo Gods' heaven...

In Security, Don't
let Perfect be the
Enemy of good

Fishing Sessions

Be sure to find Wally and sign up for 1:1 time with a speaker
If you'd like to "catch" us for a session, we are scheduled as follows:

Wednesday, May 6	
10:00am CDT	 Zero Trust: The Great Debate Lakes B Kent Agerlund • Sami Laiho
5:00pm CDT	 Behind the Scenes: Crafting a Winning Technical Session Lakes A Kent Agerlund • Sami Laiho



Don't forget to leave feedback for your speakers!

SAVE THE DATES

Oct 25-28, 2026



May 2-6, 2027



Oct 10-13, 2027



Extended Q&A



Recast



ninjaOne®



numecent®

